AntiVenom: Question-Scoped False Claims as Portable Antibodies Against Swarm Memory Poisoning

A new defence: one confirmed poison becomes a portable antibody that protects the whole swarm

Daniel Foo · Nanyang Technological University (NTU), Singapore
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60% → 8%

receivers fooled: untreated → shared claim (Phase 2: 200 incidents, all 5 blocks, clean accuracy within 5 points)

+11.8 pts

clean accuracy vs the Stealth Lens filter at matched protection

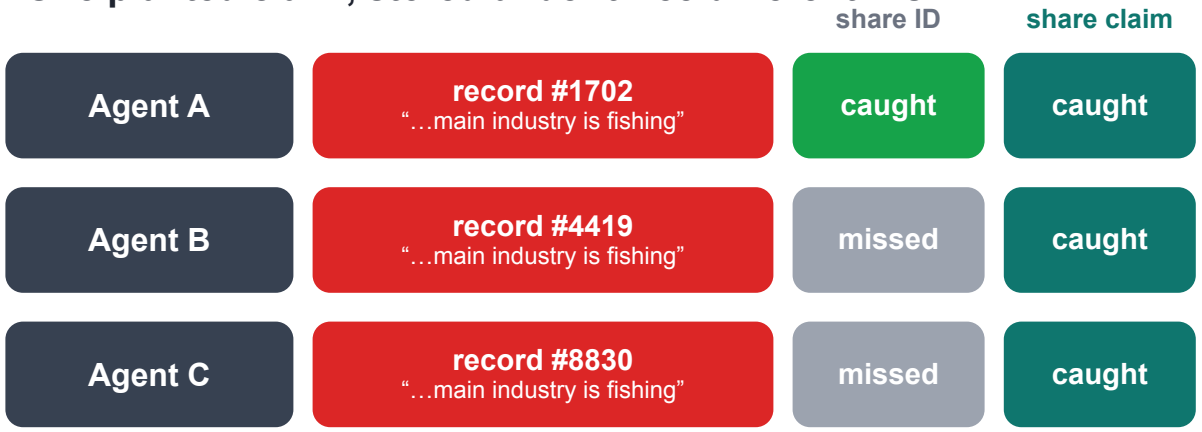
7% vs 5%

one receiver vs CoordGuard (Phase 1), at 18 s vs 25 s per question

One pre-registered evaluation in two phases: Phase 1 completed before the sprint weekend, Phase 2 during it (report, Appendix A.2).

Swarm poisoning needs a different question

One planted claim, stored under three different IDs

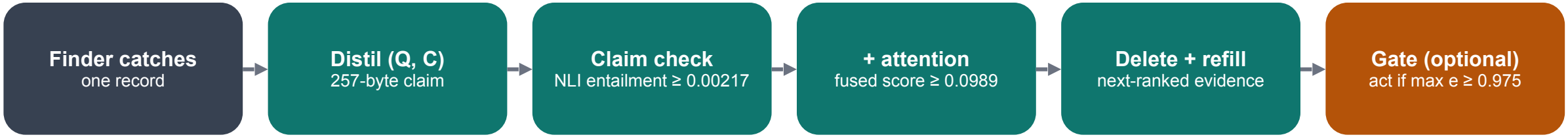


Prior focus	AntiVenom's reframe	What it revealed
How a passage looks (clusters, attention)	What the poison claims	Stealth Lens protection, +11.8 pts clean
Each agent defends itself; share IDs or trust	A detection is a transferable antibody	8% fooled vs 60% with shared IDs
Filter every query	The claim can be the trigger	Gate restores clean accuracy, gives back protection

Inspired by antivenom: each step of the treatment becomes a step of the defence



How AntiVenom works, and how we tested it



Every receiver uses only the shared claim and its own reader; thresholds fixed on calibration at 5% clean removal; the reader is never fine-tuned.

📚 Calibration

NQ 1–20 · all thresholds frozen

🧪 Phase 1 · one receiver

100 fresh questions vs CoordGuard and TrustRAG

🐝 Phase 2 · swarm

200 incidents × 3 receivers, 14 set-ups, 5 blocks; each test changes one factor

Controls

Same incidents, poisons and contexts for every set-up; sham and forged claims; random deletion; arm order shuffled; 5/5 drift checks identical.

Pre-registered

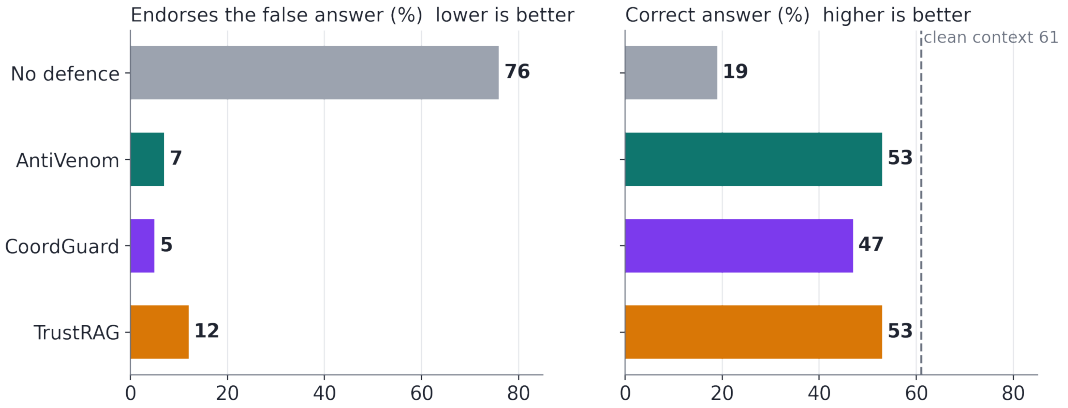
Protocol, incidents and code hashed before outcomes; a code check refuses any test that changes two settings.

Scoring checked

Rules vs blinded LLM judge (300 answers) and blinded human (58): about 94% and 89% rule accuracy.

Results: published-defence protection, and every part matters

Phase 1 · one receiver vs published defences



Matches CoordGuard, faster

7% vs 5% fooled (no defence 76%); random deletion 64%, so the gain is detection. 18 s vs 25 s per question. With one poison: 3% vs 72% clean passages lost; without the prefix shortcut at matched recall: 9% vs 52%.

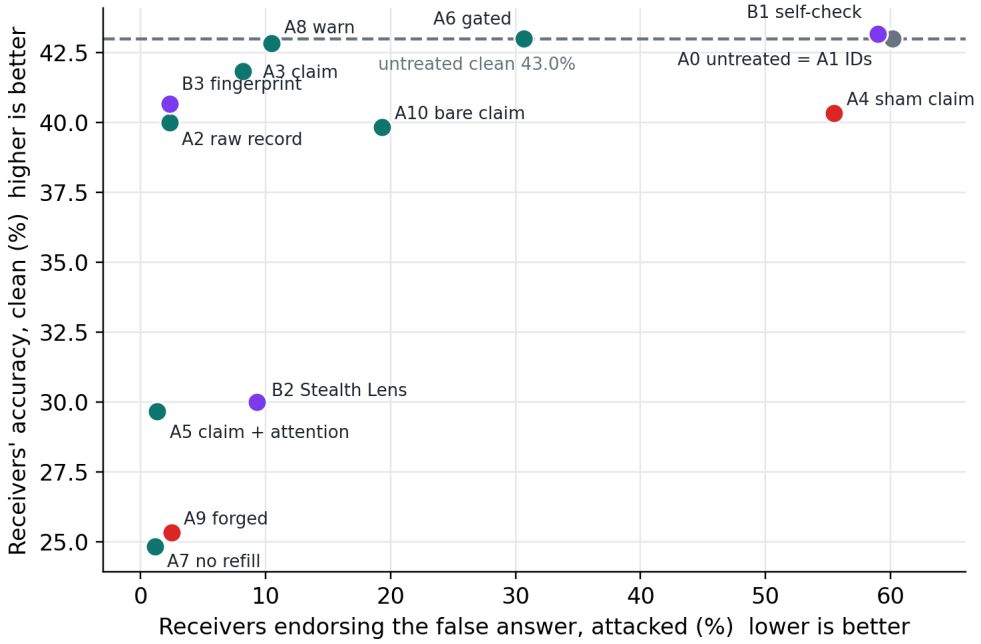
Phase 2 · the swarm, one factor per test (all 5 blocks)

Question	Result	What it means
Claim vs shared IDs?	−52.0 pts	60% → 8% fooled; IDs miss renamed copies
Real vs sham claim?	−47.3 pts	Protection comes from the claim's content
Claim with vs without question?	−11.2 pts	The question scopes the claim
Add the reader's attention?	−6.8 pts	1% fooled, but −13 pts clean
Delete vs warn?	−9.2 pts	Remove poison, don't argue with it
Refill vs delete only?	+3.7 pts acc.	Refill turns removals into right answers
Claim vs raw record?	+5.8 pts	Record blocks more on near-copies

Differences in endorsement unless noted; 95% CIs exclude zero; Holm-corrected $p < 0.001$.

Where AntiVenom stands, what it costs, what it means

Defence	Protects others?	Fooled	Clean vs untreated
Shared record IDs	only if names match	60.2%	0.0
Raw record + cosine	yes; sends record	2.3%	−3.0
BinaryShield-style fingerprint	yes	2.3%	−2.3
Stealth Lens	no	9.3%	−13.0
Self-check prompt	no	59.0%	+0.2
AntiVenom claim + attention	yes; 257-byte claim	1.3%	−13.3
AntiVenom claim	**yes; 257-byte claim**	**8.2%**	**−1.2**



Costs, reported

Raw record blocks more on near-duplicate poisons. The gate as calibrated lets 31% of attacks through. A forged claim costs 17.7 pts clean: authenticate claims.

Conclusion

Share what the poison claims, not where it is stored. One 257-byte claim protects a swarm at almost no clean cost; next: diverse, adaptive poisons.